

RUTHERGLEN, Australia

WMO: 958370

Lat: 36.1047S

Lon: 146.5094E

Elev: 574

StdP: 14.39

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.0	29.8	24.6	19.1	53.7	27.1	21.5	46.8	22.1	52.0	18.5	50.5	0.5	100	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.3	99.8	65.9	95.2	65.4	91.7	64.5	71.8	83.6	70.1	82.6	68.5	82.0	9.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	109.7	73.8	66.6	99.9	73.6	64.3	92.2	73.3	35.9	83.6	34.4	82.4	33.1	81.3	93.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.6	14.4	DB	24.9	106.0	2.4	3.4	23.2	108.5	21.8	110.5	20.4	112.4	18.7	114.9	(4)
(5)			WB	24.7	75.3	2.5	5.0	22.9	78.8	21.4	81.7	20.0	84.5	18.2	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	74.2	72.6	66.8	58.5	51.0	46.7	45.6	46.7	50.6	55.7	63.7	68.7	(6)	
	DBStd	11.69	7.63	6.72	6.73	5.76	5.12	4.52	4.00	4.26	5.12	6.08	7.34	7.09	(7)	
	HDD50	503	0	0	0	4	49	120	146	121	51	12	0	0	(8)	
	HDD65	3320	12	11	59	207	433	549	601	567	434	296	111	40	(9)	
	CDD50	3540	749	634	519	259	81	20	10	20	68	189	411	580	(10)	
	CDD65	882	297	225	113	11	0	0	0	0	1	8	72	155	(11)	
	CDH74	13293	4351	2947	1636	242	4	0	0	0	16	279	1292	2526	(12)	
	CDH80	6303	2382	1433	651	38	0	0	0	0	1	76	531	1191	(13)	
(14) Wind	WSAvg	5.6	6.9	6.1	5.8	4.3	4.3	4.1	4.9	5.0	5.8	6.1	6.7	7.2	(14)	
(15) Precipitation	PrecAvg	22.8	1.7	1.2	1.3	1.7	2.4	1.9	2.6	2.4	2.2	2.2	1.7	1.8	(15)	
	PrecMax	37.6	7.6	5.8	3.9	6.3	6.4	5.7	7.0	4.8	5.2	6.0	4.2	5.7	(16)	
	PrecMin	10.3	0.0	0.0	0.0	0.2	0.2	0.2	0.3	0.3	0.3	0.1	0.0	0.0	(17)	
	PrecStd	6.9	1.7	1.3	1.0	1.3	1.7	1.3	1.7	1.2	1.2	1.4	1.2	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	106.3	103.2	96.5	84.5	72.5	64.2	61.1	67.1	76.1	88.5	95.6	102.1	(19)	
		MCWB	66.9	67.7	66.2	61.8	57.4	54.2	53.4	57.0	58.8	61.0	63.6	64.9	(20)	
	2%	DB	101.5	96.6	90.9	79.9	68.5	60.8	58.3	62.6	70.7	81.1	90.8	95.9	(21)	
		MCWB	66.5	66.8	65.4	60.8	57.2	53.3	52.3	54.5	58.6	59.7	62.7	64.2	(22)	
	5%	DB	96.4	92.4	86.9	76.5	65.4	58.5	56.3	59.6	66.7	76.2	86.3	91.1	(23)	
		MCWB	66.0	66.0	63.9	59.8	56.3	52.5	51.1	52.7	56.6	58.8	62.5	63.0	(24)	
	10%	DB	92.0	88.7	83.2	73.2	62.4	56.2	54.1	57.0	63.6	71.8	81.6	86.7	(25)	
		MCWB	64.9	65.1	62.4	58.9	54.7	51.4	49.7	51.3	55.3	57.8	61.1	62.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.7	75.3	71.8	66.6	62.1	57.8	56.7	58.5	62.9	67.1	70.5	72.4	(27)	
		MCDB	85.6	88.4	83.9	75.5	65.7	60.2	58.3	63.4	70.7	76.2	84.3	81.0	(28)	
	2%	WB	71.4	72.2	69.3	63.7	59.3	55.1	53.5	55.7	60.6	63.9	67.9	69.1	(29)	
		MCDB	85.3	83.3	79.8	72.8	64.9	58.4	56.5	60.6	67.3	73.1	79.4	80.4	(30)	
	5%	WB	69.5	70.3	67.2	62.0	57.4	53.5	51.8	53.8	58.3	61.8	65.7	66.9	(31)	
		MCDB	84.9	81.4	79.6	71.4	62.8	56.9	55.0	58.6	64.8	70.4	77.2	81.1	(32)	
	10%	WB	67.9	68.4	65.3	60.4	55.7	52.0	50.3	51.8	56.1	59.7	63.7	65.1	(33)	
		MCDB	85.2	81.0	78.1	70.5	61.1	55.5	53.3	56.2	62.3	68.9	75.4	81.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	30.3	28.0	28.3	27.2	22.9	19.6	17.8	20.5	24.4	27.8	29.4	30.3	(35)	
		MCDBR	35.9	32.9	32.8	32.0	26.8	23.5	21.1	25.4	29.8	36.8	36.2	36.6	(36)	
		MCWBR	10.5	11.3	12.7	15.4	16.3	16.5	15.3	17.7	18.6	18.1	13.1	12.3	(37)	
	5% WB	MCDBR	25.5	22.7	25.2	26.0	21.4	19.1	17.8	21.8	25.3	26.5	25.9	26.2	(38)	
		MCWBR	9.8	10.0	11.9	13.7	14.0	14.2	13.5	15.9	17.3	16.2	11.7	11.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.355	0.348	0.320	0.314	0.301	0.292	0.288	0.297	0.326	0.327	0.345	0.345	(40)	
	taud		2.449	2.493	2.567	2.544	2.554	2.584	2.596	2.550	2.453	2.476	2.467	2.475	(41)	
	Ebn at Noon		311	306	303	286	272	266	274	288	296	309	312	315	(42)	
	Edh at Noon		38	35	30	27	23	21	22	26	33	35	37	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2438	2134	1778	1278	879	669	723	1011	1444	1908	2208	2444	(44)	
	RadStd		141	131	94	64	62	78	46	84	139	160	121	104	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+128

Nomenclature: See separate page